

# Vending Machine Management System — PRD

Product Requirements Document (PRD)

## 1. Problem Statement & Description

A vending machine needs to manage products, customer purchases, inventory, stock levels, returns, coupons, loyalty rewards, and financial performance. A basic vending machine only handles selection and payment, making it difficult for an owner to track stock, damaged or expired products, customer history, and actual profit or loss.

This command-line application provides two roles: Customer and Owner. Customers can browse products, build and edit a cart, optionally apply coupons, pay, view receipts and order history, earn and redeem loyalty points, and view a customer report. Owners can authenticate, manage stock, record damaged and expired stock, manage coupons, monitor restock alerts, view sales, and generate reports.

## 2. Interface

To run the application: `python vending_machine.py`

### Customer Entry / Identity

- Customer enters both name and phone number at the start of a session.
- The phone number is the unique customer identifier; the name is stored and displayed for the customer.
- If the phone number is new, the system creates a customer record using the entered name and phone number.
- If the phone number already exists, the system retrieves the customer's stored name, loyalty points, order history, and first-time/returning status.
- No customer password is required.

### Customer Menu

- 1. View Products — Cold Beverages, Hot Beverages, Snacks
- 2. Select Product
- 3. Add to Cart
- 4. View Cart
- 5. Edit Cart Item — change quantity / remove item
- 6. Apply / Remove Coupon
- 7. View Loyalty Points
- 8. Make Payment
- 9. View Receipt
- 10. View Order History
- 11. View Customer Report
- 12. Exit

### Owner Login / Password Setup

- The first time the owner accesses the system, the owner must create a password and confirm it before the Owner Menu is available.
- On later sessions, the owner enters the saved password to access owner functions.
- No Owner ID is required.

- An incorrect password denies access to owner functions.
- Logout returns the application to the main entry screen.

#### First run:

```
Create Owner Password: *****
Confirm Owner Password: *****
Owner password created successfully.
```

#### Later login:

```
Enter Owner Password: *****
Access granted.
```

### Owner Menu

- 1. Add Stock
- 2. Remove Stock — Damaged / Expired / Package Damage
- 3. View Inventory
- 4. View Sales
- 5. View Reports
- 6. Manage Coupons
- 7. View Restock Alerts
- 8. Logout

## 3. Functional Requirements

### FR 1: Customer Product Selection & Cart

- Display products under Cold Beverages, Hot Beverages, and Snacks with product name, selling price, available sellable quantity, and stock status.
- Customer can select a product and quantity and add it to the cart only when sufficient sellable stock exists.
- Customer can view item quantity, unit price, item total, and cart total.
- Customer can increase or decrease a cart quantity or remove an item.
- Increasing quantity must re-check current stock.
- If stock is at or below the customer warning threshold, show a message such as 'Only 2 left!' without blocking a valid purchase.

```
Select Category: Snacks
Product: Chips
Price: Rs 40
Available: 3
Only 3 left!
Enter quantity: 2
Added to cart.
Cart Total: Rs 80
```

### FR 2: Coupons & Loyalty

- Coupon use is optional. Before payment, the system must ask: 'Would you like to apply a coupon? (Y/N)'.
- If the customer selects Yes, display the coupons available to that customer with the coupon code, percentage, and minimum-order requirement.
- If the current cart total is below a coupon's minimum order, do not apply the coupon and show the exact amount still required.

- Default coupons: WELCOME40 = 40% off the first order; SAVE20 = 20% off orders of Rs 500 or more; SAVE10 = 10% off orders of Rs 1000 or more; SAVE5 = 5% off orders of Rs 2000 or more.
- WELCOME40 is available only for a first-time customer and can be used only once.
- The customer can choose 0/no coupon and continue to payment, or remove a selected coupon before payment.
- The owner can create, update, enable/disable, and remove coupons and configure their rules.
- Completed purchases earn loyalty points using the configured earning rule, for example 1 point per Rs 10 spent.
- Loyalty points persist with the customer's phone number and can be viewed at any time.
- The customer may redeem eligible loyalty points at checkout according to the configured redemption rule. The configured rule for combining loyalty redemption with a coupon must be enforced.

```

Cart Total: Rs 420
Would you like to apply a coupon? (Y/N): Y
Available Coupons:
1. SAVE20 - 20% OFF on orders of Rs 500 or more
2. SAVE10 - 10% OFF on orders of Rs 1000 or more
3. SAVE5 - 5% OFF on orders of Rs 2000 or more
SAVE20 requires Rs 500.
Add Rs 80 more to your cart to use SAVE20.
Choose a coupon or 0 to continue: 0
Continuing to payment...

```

```

Cart Total: Rs 350
Would you like to apply a coupon? (Y/N): Y
Available Coupons:
1. WELCOME40 - 40% OFF your first order
Choose a coupon: WELCOME40
40% coupon applied!
Coupon saving: Rs 140
Final Total: Rs 210

```

### FR 3: Payment, Receipt & Order History

- Calculate the amount payable after the selected coupon and any permitted loyalty redemption.
- Accept payment only when the amount provided is sufficient.
- Calculate and display change when payment exceeds the amount payable.
- After successful payment, reduce sellable stock and save the completed transaction.
- Receipt must contain order ID, date/time, customer name, items, quantities, prices, applied coupon, loyalty redemption if used, amount paid, change, and final amount.
- Every completed transaction is saved against the customer's phone number.
- Customer can view previous orders and open the full receipt for a selected order.

```

Final Total: Rs 210
Enter payment: Rs 250
Payment successful!
Change: Rs 40
Order ID: ORD1007
Receipt saved to order history.

```

### FR 4: Expiry, Returns & Loss Management

- When the owner adds stock, the owner must enter the expiry date for the relevant stock batch.
- Products nearing expiry, for example within 3 days, must be flagged for the owner.
- Expired products must automatically become unavailable for sale and must be blocked during final cart/payment validation.

- A customer may return a purchased product only when the product is damaged or expired.
- The system must verify that the item belongs to a completed purchase and that the return quantity does not exceed the purchased quantity.
- For an expired-product return, the relevant expiry date must be recorded/verified.
- For a damaged-product return, the customer must provide a description of exactly what happened.
- A valid return generates a refund based on the recorded purchase.
- Returned damaged/expired products are not returned to sellable inventory.
- The returned quantity, reason, description where applicable, and resulting loss must be recorded.
- Owner stock removal must support Damaged, Expired, and Package Damage. Expired stock requires the expiry date; damaged/package-damaged stock requires a description.

Owner removes expired stock:  
 Product: Milk Coffee  
 Quantity: 5  
 Expiry Date: 08-09-2026  
 Reason: Expired  
 Stock removed from sellable inventory.  
 Expired loss recorded.

Owner removes damaged stock:  
 Product: Chips  
 Quantity: 2  
 Reason: Package Damage  
 Description: Packet was torn and the contents were exposed.  
 Stock removed from sellable inventory.  
 Package-damage loss recorded.

Customer return:  
 Order ID: ORD1007  
 Product: Chips  
 Quantity to return: 1  
 Reason: Damaged  
 Description: Packet was torn before opening.  
 Return verified.  
 Refund: Rs 40  
 Returned item is not added back to sellable stock.

## FR 5: Owner Inventory & Restock Management

- Owner must authenticate with the saved owner password before using owner functions.
- Owner can add stock with product, quantity, cost information, and expiry date.
- Owner can remove stock for Damaged, Expired, or Package Damage.
- Inventory must separately track sellable stock and removed/loss quantities.
- Each product has an owner-configurable minimum stock threshold.
- When sellable stock falls at or below the threshold, the product must appear in Restock Alerts.

Product: Cold Coffee  
 Current Stock: 5  
 Restock Threshold: 5  
 RESTOCK ALERT  
 Cold Coffee has reached its minimum stock level.

## FR 6: Sales, Coupons & Reports

- Owner can view recorded sales containing order information, products, quantities, selling prices, applied coupons, loyalty redemption, and final amounts.

- Owner can manage coupon definitions and status.
- Owner can generate Sales, Inventory, Damaged Product, Expired Product, Return, Low Stock, and Profit/Loss reports.
- Customer can view a customer report containing their purchase history and loyalty information.
- Reports must be generated from recorded transaction, inventory, return, coupon, and loyalty data.

Owner Menu

5. View Reports

Select Report:

1. Sales

2. Inventory

3. Damaged Product

4. Expired Product

5. Return

6. Low Stock

7. Profit/Loss

Choose: 6

LOW STOCK REPORT

Cold Coffee - Stock: 5 - Threshold: 5

## FR 7: Profit & Loss Calculation

- Track sales revenue, product acquisition costs, coupon reductions, refunds, damaged-product losses, expired-product losses, and package-damage losses.
- Returned damaged/expired products must be included in the recorded loss according to the project's accounting treatment.
- The system must calculate and display the resulting net profit or loss using one clearly defined accounting formula.
- The report must use recorded transaction and inventory data rather than manually entered totals.

Profit/Loss Report

Sales Revenue: Rs 500

Product Costs: Rs 280

Coupon Reductions: Rs 50

Refunds: Rs 40

Recorded Stock Losses: Rs 95

Net Result: calculated from the defined project formula

## 4. Edge Cases

| EC    | Situation                                                                   | Expected Behaviour                                                                                                                                                       |
|-------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EC 1  | Invalid product or quantity                                                 | Reject input and request a valid product and positive whole-number quantity.                                                                                             |
| EC 2  | Requested quantity exceeds sellable stock                                   | Reject request and show the available quantity.                                                                                                                          |
| EC 3  | Cart quantity exceeds current stock                                         | Reject the edit and keep the previous valid quantity.                                                                                                                    |
| EC 4  | Coupon invalid, expired, disabled, ineligible, or minimum order not reached | Reject coupon; if minimum is not reached, show the exact shortfall.                                                                                                      |
| EC 5  | Customer chooses not to use a coupon                                        | Continue directly to payment without applying a coupon.                                                                                                                  |
| EC 6  | Payment is less than final total                                            | Reject payment and request sufficient payment.                                                                                                                           |
| EC 7  | Product expires before final payment                                        | Block the product from sale after the final expiry check.                                                                                                                |
| EC 8  | Invalid return                                                              | Reject return when purchase cannot be verified, quantity is too high, or reason is not damaged/expired.                                                                  |
| EC 9  | Owner removes more stock than available                                     | Reject removal and leave inventory unchanged.                                                                                                                            |
| EC 10 | Owner enters invalid/mismatched password during setup or login              | Reject access; during setup require matching password confirmation.                                                                                                      |
| EC 11 | Stock reaches restock threshold                                             | Create/display the restock alert.                                                                                                                                        |
| EC 12 | Customer has no previous orders or loyalty points                           | Show first-time status and an appropriate zero/empty history and points state.                                                                                           |
| EC 13 | Checkout with an empty cart                                                 | Reject checkout and inform the customer the cart is empty; return to the menu without proceeding to payment.                                                             |
| EC 14 | Redeeming more loyalty points than available                                | Reject the redemption request and show the customer's actual available point balance.                                                                                    |
| EC 15 | Invalid coupon + loyalty combination                                        | Reject the combination per the configured rule and inform the customer which one (coupon or loyalty redemption) cannot be applied together.                              |
| EC 16 | Returning more than the remaining eligible quantity                         | Reject the return and show the remaining quantity still eligible for return.                                                                                             |
| EC 17 | Reusing a one-time coupon                                                   | Reject the coupon and inform the customer it has already been used.                                                                                                      |
| EC 18 | Adding zero/negative stock                                                  | Reject the stock entry and request a positive whole-number quantity.                                                                                                     |
| EC 19 | Adding already-expired stock                                                | Reject the entry (or flag it as immediately expired/unsellable per the defined policy) and prevent it from entering sellable inventory with a past expiry date.          |
| EC 20 | Exact payment with Rs 0 change                                              | Accept payment, complete the transaction, and display change as Rs 0.                                                                                                    |
| EC 21 | Existing phone number entered with a different name                         | Retrieve the existing customer record by phone number per the defined name-update policy (e.g., keep the stored name or update it), without creating a duplicate record. |

| EC    | Situation                                         | Expected Behaviour                                                                                                                     |
|-------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| EC 22 | Data persistence after restarting the application | All saved data (customers, inventory, coupons, orders, loyalty points) must be reloaded correctly and remain consistent after restart. |

## 5. Test Cases

| FR     | Test                     | Input / Situation                                                                              | Expected Output                                                                       |
|--------|--------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| FR 1   | View products            | Customer selects Snacks                                                                        | Snack products show name, price, quantity, and stock status.                          |
| FR 1.1 | Low stock                | Product stock = 2                                                                              | Show 'Only 2 left!' and allow purchase if stock is available.                         |
| FR 1.2 | Edit cart                | Increase quantity beyond current stock                                                         | Edit rejected; previous valid quantity remains.                                       |
| FR 2   | Coupon shortfall         | Cart Rs 420; SAVE20 requires Rs 500                                                            | Show 'Add Rs 80 more...' and do not apply SAVE20.                                     |
| FR 2.1 | First-time coupon        | New customer selects WELCOME40 on Rs 350 cart                                                  | Apply 40%; coupon saving Rs 140; final total Rs 210.                                  |
| FR 2.2 | Returning customer       | Customer who has completed an order tries WELCOME40                                            | WELCOME40 is not accepted.                                                            |
| FR 3   | Payment                  | Final total Rs 210; payment Rs 250                                                             | Payment succeeds and Rs 40 change is displayed.                                       |
| FR 3.1 | History                  | Customer opens a previous order                                                                | Full saved receipt is displayed.                                                      |
| FR 4   | Expiry                   | Owner adds stock with expiry date; date passes                                                 | Product becomes unsellable and appears in expired tracking.                           |
| FR 4.1 | Damaged stock            | Owner removes 2 Chips; description entered                                                     | Units removed and damage description/loss recorded.                                   |
| FR 4.2 | Return                   | Customer returns purchased damaged item with description                                       | Return verified, refund issued, item not returned to sellable stock.                  |
| FR 5   | Restock                  | Stock falls to configured threshold                                                            | Product appears in Restock Alerts.                                                    |
| FR 6   | Reports                  | Owner selects Sales Report                                                                     | Recorded sales are displayed.                                                         |
| FR 7   | Profit/Loss              | Revenue, costs, coupon reductions, refunds, and losses are recorded                            | Report calculates the net result using the defined formula.                           |
| EC 1   | Invalid product/quantity | Customer enters an invalid product code or a non-numeric/negative quantity                     | Input rejected; system requests a valid product and a positive whole-number quantity. |
| EC 2   | Quantity exceeds stock   | Customer requests 10 units of a product with 4 sellable units available                        | Request rejected; system shows the available quantity (4).                            |
| EC 3   | Cart edit exceeds stock  | Customer tries to increase a cart item's quantity beyond current sellable stock                | Edit rejected; the cart keeps the previous valid quantity.                            |
| EC 4   | Coupon not eligible      | Customer tries to apply an expired, disabled, or ineligible coupon, or one below minimum order | Coupon rejected; if minimum order not reached, exact shortfall amount is shown.       |
| EC 5   | Decline coupon           | Customer answers 'N' to the apply-coupon prompt                                                | System proceeds directly to payment without applying any coupon.                      |
| EC 6   | Insufficient payment     | Final total Rs 210; customer enters payment of Rs 150                                          | Payment rejected; system requests sufficient payment.                                 |
| EC 7   | Expiry at final check    | Product expires between cart addition and final payment confirmation                           | Product is blocked from sale at the final expiry check before payment completes.      |

| FR    | Test                            | Input / Situation                                                                                                   | Expected Output                                                                                                               |
|-------|---------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| EC 8  | Invalid return                  | Customer attempts to return an item from an unverifiable order, an excess quantity, or a non-damaged/expired reason | Return rejected with the specific reason (purchase not verified / quantity too high / invalid reason).                        |
| EC 9  | Excess stock removal            | Owner attempts to remove 20 units of Damaged stock when only 8 are in inventory                                     | Removal rejected; inventory quantities remain unchanged.                                                                      |
| EC 10 | Password mismatch               | Owner enters mismatched password/confirmation during setup, or a wrong password during login                        | Access denied; setup requires the confirmation to match before a password is saved.                                           |
| EC 11 | Restock threshold reached       | Sellable stock of Cold Coffee drops to 5, equal to its configured threshold                                         | Cold Coffee is created/displayed in the Restock Alerts list.                                                                  |
| EC 12 | First-time customer, no history | A brand-new phone number is entered at session start                                                                | System shows first-time status with zero order history and zero loyalty points.                                               |
| EC 13 | Empty cart checkout             | Customer selects Make Payment with no items in cart                                                                 | System rejects checkout, shows 'Your cart is empty', and returns to the customer menu.                                        |
| EC 14 | Over-redeem loyalty points      | Customer has 30 points and requests to redeem 50                                                                    | Redemption rejected; system shows available balance of 30 points.                                                             |
| EC 15 | Invalid coupon + loyalty combo  | Customer applies SAVE10 and then tries to also redeem loyalty points where combination is disallowed                | System rejects the second selection and states that a coupon and loyalty redemption cannot be combined.                       |
| EC 16 | Over-return quantity            | Customer purchased 2 Chips, already returned 1, requests to return 2 more                                           | Return rejected; system shows 1 unit remains eligible for return.                                                             |
| EC 17 | Reuse one-time coupon           | Returning customer who already used WELCOME40 attempts to apply it again                                            | Coupon rejected with message that WELCOME40 has already been used.                                                            |
| EC 18 | Zero/negative stock add         | Owner attempts to add stock with quantity 0 or -5                                                                   | Entry rejected; system requests a positive whole-number quantity.                                                             |
| EC 19 | Add already-expired stock       | Owner adds stock with an expiry date earlier than the current date                                                  | Entry rejected (or flagged as immediately expired per policy) and not added to sellable inventory.                            |
| EC 20 | Exact payment                   | Final total Rs 150; payment Rs 150                                                                                  | Payment succeeds; change displayed as Rs 0.                                                                                   |
| EC 21 | Same phone, different name      | Existing phone number entered with a new/different name at session start                                            | System retrieves the existing record by phone number and applies the defined name-update policy without creating a duplicate. |
| EC 22 | Persistence after restart       | Application is closed and restarted after transactions, stock changes, and coupon usage                             | All customer, inventory, coupon, order, and loyalty data are reloaded correctly and match the pre-restart state.              |